22 Find the coordinates of the turning point on the curve with equation $y = 9 + 18x - 3x^2$
You must show all your working.

$y = 9 + 18x - 3x^2$

$= -3(x^2 - 6x) + 9$

$= -3[(x - 3)^2 - 9] + 9$

$= -3(x - 3)^2 + 36$

Maximum at $x = 3$, $y = 36$.

1 mark
factor out −3

1 mark
$-3(x-3)^2+36$

1 mark
start completing square

1 mark
turning point (3,36)

(3 , 36)

(Total for Question 22 is 4 marks)

TOTAL FOR PAPER IS 80 MARKS

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